
How much search budget a target fidelity actually needs

Answering last checkpoint's action item: two boundary runs, decided on validation only.

Known before: physics-informed proposals beat random and greedy circuit search in all seven tested conditions; the closed-loop bonus was not general.

Measured now: on the Ising chain the open loop meets the rule at budget 6 (12/12) while the closed loop does not (8/12); raising the XXZ budget from 8 to 10 moved the closed loop from 9/12 to 5/12.

Not claimed: no single minimum budget, no interval, and nothing about the held-out test set, which was never evaluated in this work.

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Project: llm-vqc · Previous checkpoint: one-factor robustness stress test · New model calls: 138 · candidate circuits trained: 408 · Measured cost at list price: USD 0.29 (hard cap USD 2.00, authorised in advance)

llm-vqc

What budget reaches a required accuracy, not which method wins

Every earlier comparison ranked methods at one fixed budget; that is not the operational question.

For a target validation trash fidelity, what is the smallest candidate budget that reaches it in at least 10 of the same 12 paired seeds?

- **Claim under test (EXPERIMENT_CHECK_REQUIRED before this run):** a budget below 8 suffices on the Ising chain, and the XXZ cell that fell one seed short at budget 8 is rescued by more budget.
- Alternative that fits the same evidence (MODEL_INFERENCE): attainment counts near a threshold are dominated by run-to-run variation, so a budget change moves the count without moving the underlying fidelity distribution much.
- Both were separable in advance, because each budget is run as its own independent policy rather than as a prefix of a larger run.
- A result could exist under a false claim only if the counts were monotone in budget by construction. They are not: the runs are independent, so this was a genuine test.

The endpoint is a count of seeds over a fixed threshold, not a mean fidelity.

Next: What was already settled, and what this adds.

Settled: semantics helps. Unsettled: how much search it needs

Position relative to the immediately preceding checkpoint of this project.

Settled (CONFIRMED, measured)

- Physics-informed proposals beat non-semantic search in all seven previously tested conditions.
- The result survives changes of budget, width, Hamiltonian and model.

Already shown fragile (measured)

- The extra gain from a closed feedback loop is not general; it changes with every factor tested.
- Invalid proposals become random draws, entangling validity with architecture quality.

The gap this run addresses

- No budget between 4 and 8 had ever been run.
- One cell sat exactly one seed short of the rule and had never been retried at a larger budget.

Novelty: recombined

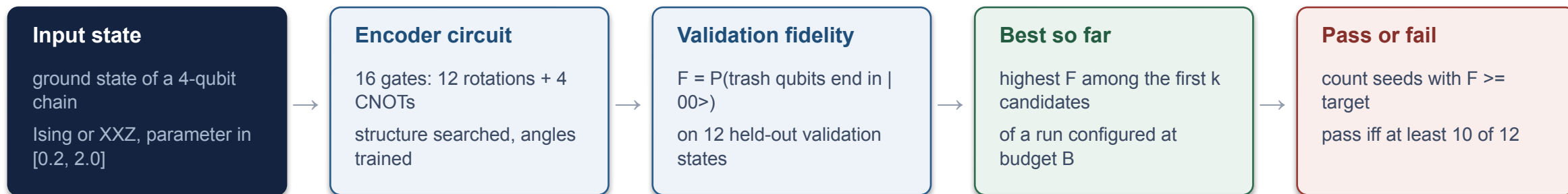
- The methods, prompts, trainer and seeds are unchanged; only the budget moves.
- Closest prior work is this project's own previous checkpoint, reused here as controls.

Next: Every term the rest of the deck uses.

From a ground state to a single pass-or-fail count

Each arrow is a fixed rule, written down before the runs; nothing below was chosen after seeing a result.

Budget B candidate circuits trained and evaluated per seed - not tokens, not model calls, not wall-clock time



A mean above the target is not a pass, and equality with the target counts as a hit.

Next: The judging rules, and the one that was deliberately changed.

The judging rules were frozen and committed before any run

Fixed in the repository before the first new candidate circuit was generated.

Rule	Fixed as
Endpoint	Validation trash fidelity; never test
Targets	0.95 and 0.99, both reported, neither revisable
Seeds	The same 12 paired seeds, 0-11, all reported
Pass rule	At least 10 of 12; equality counts
Admissible budgets	Even budgets 4, 6, 8, 10, 16 only
Budget split	Half exploration, half refinement, always
Inherited unchanged	Model, prompts, schema, repair policy, gates, trainer, splits, selection
Cost	Refuses to call without an explicit cap; USD 2.00
Reported as	Verified pass / verified fail / unverified

- One deliberate change from the action item: no interval for the minimum is reported.
- An interval would assume attainment never disappears as the budget grows. That assumption is testable, and slide 7 shows it failing, so the three-way report replaces it.

Does not show: The table does not show effect sizes or significance tests; the endpoint here is a count, and no paired statistic is claimed from it.

Next: What was actually executed, and what was reused.

Two new runs, seven reused conditions, no test evaluation

Existing logs were re-analysed first; new calls were made only at the two unresolved boundaries.

Reused (no model calls)

- Seven earlier conditions, 2,880 candidate records.
- Re-analysis reproduces the earlier audit byte for byte (CONFIRMED by directory comparison).

Run now

- Ising chain at budget 6, all four methods, same seeds.
- XXZ chain at budget 10, closed loop only, same seeds.
- 408 candidate circuits trained and evaluated.

Checks that passed

- Seed completeness, candidate order, finite fidelity, selection equals validation max.
- A machine check refuses any cell that moves more than the budget away from its declared anchor.
- 603 automated tests pass, 46 of them new.

Not verified

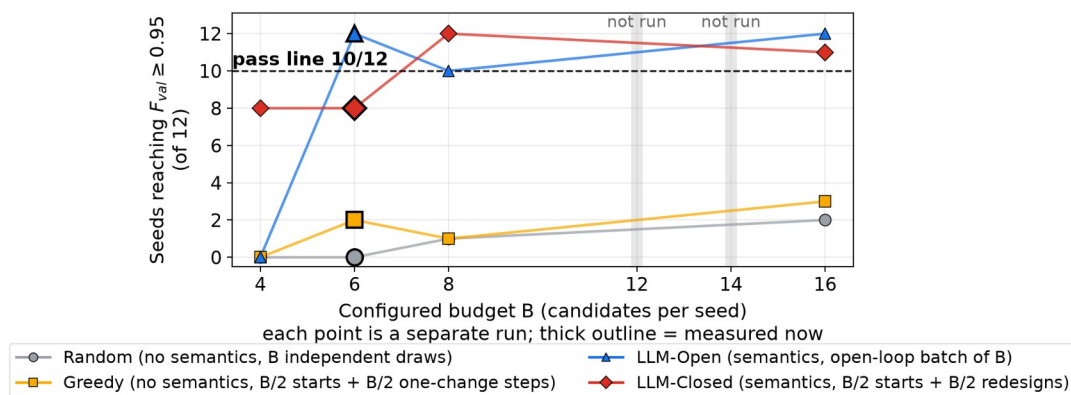
- No confirmatory evaluation on any held-out set exists, by design.
- One run was interrupted by a network failure; the resume paid nothing for stored seeds.

Next: The main evidence.

Runner, analysis, figures, report and this deck are generated by checked-in scripts; the reproduction commands are in the report.

The open loop clears the bar at a budget the closed loop cannot

Using chain, target 0.95, four independently executed budgets, the same 12 paired seeds.



Metric: validation trash fidelity (never test). 12 paired seeds (0-11). Pass = at least 10/12 seeds at or above the target. Runs at different B are separate policies and are never concatenated.

x: configured budget B , candidate circuits per seed. *y*: how many of the 12 paired seeds finished at or above validation fidelity 0.95. Dashed line: the 10-of-12 rule. Thick outline: measured here. Grey bands: budgets not run. Grey random, amber greedy, blue open loop, red closed loop.

Does not show: No error bars - the quantity is a count over a fixed seed set, not an average. Nothing about budgets 12 or 14, the second target, other widths, or held-out test performance.

Next: What this does and does not mean.

Measured

- Budget 6: open 12/12 passes, closed 8/12 fails.
- Same budget and seeds: random 0/12, greedy 2/12.

Counts are not monotone

- Open over budgets 4, 6, 8, 16: 0, 12, 10, 12.
- Closed over the same budgets: 8, 8, 12, 11.

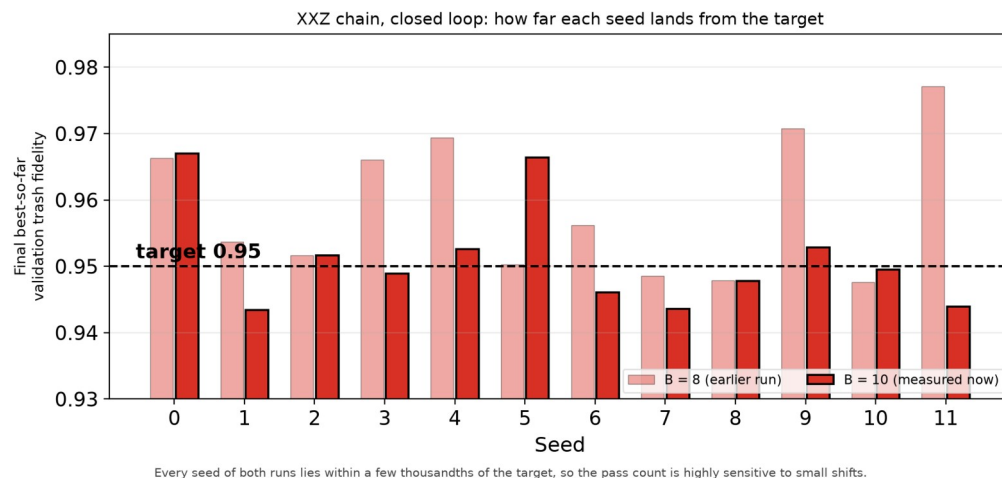
Null result

- Target 0.99: 0 of 12 in both new runs.
- Not evidence that it is unreachable.

Generated from the committed validation-only result tables by the checked-in figure script.

The XXZ count halved while the fidelities barely moved

Closed loop at the XXZ anchor, budget 8 against budget 10, per seed, same 12 paired seeds.



x: seed. y: the final best-so-far validation trash fidelity that seed reached - pale for the earlier budget-8 run, solid for the budget-10 run measured here. Dashed line: the 0.95 target. Bars are single measured values, not averages.

Does not show: Not the trajectory within a run, not the rejected candidates, and not whether a third run would land the same way.

Observed (measured)

- Seeds at the target: 9 to 5 of 12.
- Mean fidelity moved only -0.0076; 17 of 24 results sit within 0.01 of it.

Interpretation (inferred)

- Near a threshold inside the distribution, the count is a knife-edge statistic.
- Only the budget was varied, so no cause is isolated.

Not supported

- That more budget harms the closed loop.
- That the closed loop is generally worse.

Supports: the budget boundary is real and the count is threshold-sensitive. **Does not support:** any causal story about feedback.

Next: What would have to be true for this to be wrong.

What bounds this result, and what is recorded but unresolved

Stated before any of it is used to argue for a next step.

What would overturn it

- A repeat landing on the other side of the threshold; margins are thousandths wide.
- A different target, which would reshuffle every count.

Validity range

- 4 qubits, noiseless simulation, one circuit-capacity rule, one model snapshot, one XXZ anchor.
- 12 paired seeds; one open-loop pool shared across them, so 12 successes are not 12 generations.

Negative and null results kept

- The closed loop fails the rule at budget 6.
- The XXZ boundary probe did not rescue the cell.
- Target 0.99: 0 of 12 everywhere, unresolved.

Left unresolved on purpose

- 6 and 8 qubits and the alternative model were not probed; every method sits at 0-3 of 12 there.
- A blind sweep cannot separate invalid generation from too little search or capacity.
- Random fallbacks stay in the score: 12 of 408 evaluated candidates.

Open question: Attainment counts are not monotone in budget, so no statement of the form 'the minimum lies between X and Y' is supported by these runs.

Next: What needs deciding.

Spend more on resolution, or diagnose why 0.99 is out of reach

Both new cells are complete; the next move needs a decision, not more of the same.

What the evidence supports

- Ising chain, target 0.95: smallest verified passing budget is 6 for the open loop, 8 for the closed loop; budget 4 fails for both.
- XXZ chain, closed loop: no verified passing budget; 8 and 10 are both verified failures (measured).
- Target 0.99 unresolved everywhere, both new cells included (null result).
- New work: 138 model calls, USD 0.29 at list prices, under a USD 2.00 cap.

Decisions needed from you

- **D1 Is the budget boundary worth more spend?**
 - (a) repeat both cells with fresh generation seeds - turns a knife-edge count into a distribution
 - (b) stop refining; treat budget 6 open-loop as the working answer - free, but rests on one run
 - Recommended (a), Ising cell only; moderate confidence.
- **D2 What to do about the 0.99 target**
 - (a) diagnose circuit capacity and training length instead of search
 - (b) record 0.99 as out of scope
 - Recommended (a), moderate confidence; a budget sweep cannot separate the causes.
- **D3 Any confirmatory read of held-out data**
 - Recommended: not yet, high confidence.

Proposed next step

- On D1(a): repeat the Ising budget-6 cell with fresh generation seeds; report both runs together.
- On D2(a): write the capacity-versus-search protocol and its cost cap, then stop for review.
- A yes to either commits to a new protocol document and a new cost cap before anything runs.
- Already committed: report, raw logs with hashes, and the reproduction commands.

Recommendation and decision are recorded separately in the research log; nothing above is approved.